PADHAKU
Exploration · Class 9 Science



CHAPTER 13

Earth as a System: Energy, Matter, and Life

NCERT Exercise Solutions

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Try each question yourself first, then check the answers. answers are samples: write your own.

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🤔 Think It Over

1 **How does the warming of Arabian Sea water affect the southwest monsoon in India?**

✔ ANSWER

Warmer sea water evaporates more, so more moisture enters the air.

This causes ups and downs (fluctuations) in the southwest monsoon.

Rainfall becomes uneven: **floods in some regions and drought in others**.

2 **If a large forest is cleared, how can that affect the flow of a river in that area?**

✔ ANSWER

Without tree roots, soil is washed away (soil erosion) and silt fills the river.

Less water soaks into the ground, so rain runs off quickly → **floods in the rainy season**.

Less groundwater and less transpiration (less local rain) → **the river may run low or dry in dry months**.

3 **What might happen to coastal cities in India if glaciers and polar ice keep melting faster?**

✔ ANSWER

Melted ice adds water to rivers and oceans, so **sea level rises**.

Low-lying coastal cities like Mumbai and Chennai may face flooding, land loss and damage to homes and habitats.

4 **How would increasing carbon dioxide levels in the atmosphere affect the ocean plankton?**

✔ ANSWER

More CO_2 dissolves in the ocean and makes sea water **more acidic**.

This can harm tiny plankton (and shell-forming organisms, coral reefs), disturbing the marine food chain.

🧪 Activity 13.1: Let us explore

1 **Observe Fig. 13.1 (mountains, snow, lake, grass, sheep, shepherd). Identify one example of each sphere.**

✔ ANSWER

- Geosphere: rocks, mountains, soil
- Hydrosphere: the lake
- Cryosphere: snow on the mountain tops
- Atmosphere: the air, sky and clouds
- Biosphere: grass, sheep, the shepherd

2 **How does snow (cryosphere) eventually become part of the lake (hydrosphere)?**

✔ ANSWER

In summer the Sun's heat melts the snow.

The melt water flows down the slopes as streams and collects in the lake.

3 **If there is less snowfall during winters for a few years, how would this affect the lake's level and the grass available for the sheep?**

✔ ANSWER

Less snow → less melt water in summer → **the lake level falls.**

Less water for the land → **less grass grows**, so the sheep get less food.

4 **Discuss and write how all the spheres are interconnected, and how a disturbance in one can lead to changes in others.**

💡 SAMPLE ANSWER

The Sun's heat melts snow (cryosphere) into the lake (hydrosphere); water evaporates into the air (atmosphere) and falls as rain/snow; water and soil (geosphere) let grass grow, which feeds sheep and people (biosphere).

So a change in one sphere spreads to others, e.g. less snow → less lake water → less grass → fewer sheep can be fed.

🤔 Think as a Scientist · 13.1 Uneven Heating of the Earth

? **Estimate how much of the Earth's surface would need to be covered with solar panels to supply all the electric power India uses today.**

💡 SAMPLE ANSWER

Assumptions (approximate, check latest numbers on the internet):

- India uses about 1,800 billion kWh of electricity a year → average power ≈ 1,800 × 10^9 kWh ÷ 8760 h ≈ **2 × 10^8 kW**.
- Average sunlight over day and night ≈ 5 kWh per m^2 per day ≈ 0.2 kW m^{-2}.
- Panels convert about 20% → 0.2 × 0.2 = 0.04 kW per m^2.

Area = 2 × 10^8 kW ÷ 0.04 kW m^{-2} = 5 × 10^9 m^2 ≈ **5,000 km^2** (about 70 km × 70 km).

The Thar desert is about 2,00,000 km^2, so only about **2–3%** of it would be enough.

Activity 13.2: Let us find out · 13.1.1 Interaction of solar radiation on the Earth's surface

1 **Complete Table 13.1 (albedo of surfaces) using authentic sources.**

S. No.	Material	Albedo
1	Snow	0.80 – 0.90
2	Ice	0.50 – 0.70
3	Crushed rock	0.25 – 0.30
4	Light coloured soil	?
5	Black soil	?
6	Ocean water	?

💡 **SAMPLE ANSWER**

Approximate values (sources differ a little):

S. No.	Material	Albedo
4	Light coloured soil	about 0.25 – 0.40
5	Black soil	about 0.05 – 0.15
6	Ocean water	about 0.06 – 0.10

Snow and ice reflect most sunlight (high albedo); black soil and ocean water absorb most of it (low albedo), so they stay warmer.

⭐ Pause and Ponder · 13.1.3 Role of the atmosphere

1 **Visit https://phet.colorado.edu/en/simulations/greenhouse-effect and study the effect of the concentration of greenhouse gas on surface temperature.**

💡 SAMPLE ANSWER

Observation: as the greenhouse gas concentration is increased, more infrared (heat) from the surface is trapped and **the surface temperature rises**; with fewer greenhouse gases more heat escapes to space and the surface cools.

Conclusion: some greenhouse gases keep the Earth warm enough for life, but too much causes global warming.

⭐ Pause and Ponder · 13.2.3 Ocean currents

2 **How does the cool mountain breeze benefit agriculture activity, particularly the crops and soil?**

✔ ANSWER

At night the cool mountain breeze flows down into the valley and lowers the temperature.

It helps regulate temperature and keeps moisture in the soil (less evaporation, dew forms).

This reduces heat stress on crops and supports **soil and crop health**.

3 **What happens to the warm surface water from the equator as it travels toward the poles? What impact does this movement have on the area?**

✔ ANSWER

The warm water gives its heat to the air and **cools down**; colder, denser water sinks and slowly flows back to the equator at deeper levels.

Impact: it warms the air of colder regions and makes their climate milder, e.g. the North Atlantic Drift keeps many ports of north-western Europe **ice-free in winter**, helping trade. Currents also carry nutrients that support marine life.

⭐ Pause and Ponder · 13.3.2 Carbon cycle

4 **The CO_2 dissolved in the ocean is disturbed when the global temperature increases. What will happen to marine life?**

✔ ANSWER

Warmer water holds less dissolved CO_2, so the ocean absorbs less CO_2 and more stays in the air (more warming).

With extra CO_2, sea water becomes more acidic; less carbonate is available for shells.

So **phytoplankton, shell-forming organisms and coral reefs are harmed**, and the whole marine food chain (fish, etc.) is affected.

Pause and Ponder · 13.3.4 Oxygen cycle

5 **What would happen to plants and animals on Earth if the biogeochemical cycles were disrupted and stopped? Explain with examples.**

✔ ANSWER

Nutrients would not be recycled, so they would run out and life could not continue.

- Water cycle stops → no rain; plants dry up, animals have no drinking water.
- Carbon/oxygen cycle stops → plants get no CO_2 for photosynthesis, so no food and no new O_2 for respiration.
- Nitrogen cycle stops → plants cannot make proteins; animals that eat them also suffer.
- Dead bodies and wastes would pile up without decomposition.

What if ... · 13.4 Human Impact on Earth's Processes

? **If photosynthesis stopped, what would happen on the Earth?**

✔ ANSWER

- No new food (glucose) would be made, so food chains would collapse; plants, then animals and humans, would starve.
- No oxygen would be returned to the air; O_2 would slowly fall and CO_2 would keep rising.
- More CO_2 would increase global warming. Life on Earth could not survive for long.

Pause and Ponder · 13.4 Human Impact on Earth's Processes

6 **Discuss how human activities increase the concentration of greenhouse gases in the atmosphere. What would you do as an individual to reduce the emission of greenhouse gas?**

💡 SAMPLE ANSWER

How humans increase them:

- Burning coal, petrol, diesel and gas for electricity, vehicles and industry releases CO_2.
- Cutting forests (deforestation) means less CO_2 is absorbed.
- Farming, cattle, rice fields and garbage dumps release methane; fertilisers add nitrogen gases.

What I can do:

- Switch off lights and fans when not needed; use LED bulbs.
- Walk, cycle or use public transport; plant and care for trees.
- Reduce, reuse, recycle; avoid wasting food and water; support solar energy.

Revise, Reflect, Refine

1 **Choose the most appropriate option to describe the role of biogeochemical cycles in an ecosystem.**

(i) To provide food directly to all organisms.

(ii) To recycle essential nutrients between biotic and abiotic components.

(iii) To create new elements for use by living things.

(iv) To remove pollutants and toxins from the organism.

✔ ANSWER

(ii) To recycle essential nutrients between biotic and abiotic components.

Cycles move matter like C, N, O and water between living and non-living parts; they do not create new elements.

2 **Which of the following is primarily responsible for warming of the Earth?**

(i) Solar radiation is immediately absorbed by carbon dioxide, which then releases it as heat.

(ii) The atmosphere's tiny particles absorb incoming solar radiation, which directly heats the Earth.

(iii) The Earth's surface absorbs solar radiation, which is then re-radiated and trapped by greenhouse gases.

(iv) The Earth's environment is heated only by the solar radiation reflected by the clouds.

✔ ANSWER

(iii)

The surface absorbs sunlight, re-radiates it as infrared (heat), and greenhouse gases (CO_2, CH_4, water vapour) trap part of this heat.

3 **Explain how climate change affects the water cycle. Illustrate with examples.**

✔ ANSWER

- A warmer atmosphere holds more moisture → **heavier rain** in some areas (intense monsoons) and **droughts** elsewhere.
- Warmer Arabian Sea → more evaporation → changes in the southwest monsoon, floods in some regions, drought in others.

- Melting glaciers add more water to rivers and raise sea level, threatening coastal cities like Mumbai and Chennai.
- Sudden heavy bursts of rain → more run-off, soil erosion and less infiltration → less groundwater recharge, so farming becomes hard in dry months.

4 **Describe how albedo affects the Earth's surface temperature and its climate.**

✔ ANSWER

Albedo = fraction of sunlight a surface reflects.

- High albedo surfaces (snow, ice ~0.8–0.9) reflect most sunlight and stay cool → polar regions are very cold.
- Low albedo surfaces (black soil, ocean water, dark roads) absorb more and become warmer.
- When snow and ice melt or forests are cleared, albedo changes; less ice means more absorption and more warming. So albedo controls how much the Earth heats up and shapes local and global climate.

5 **How are mountain and valley breezes formed? Suppose there are two mountains, one covered with grass and another with barren rocks; would the temperature of the two mountain breezes be different? If so, how?**

💡 SAMPLE ANSWER

Valley breeze (day): sunny slopes heat faster than the valley floor; warm air over the slopes rises (low pressure) and cooler valley air moves **up** the slopes.

Mountain breeze (night): slopes lose heat faster and become cooler; cool, dense air flows **down** into the valley.

Yes, they would differ. Barren rock heats up and cools down more quickly and re-radiates heat fast, so at night its slopes get very cold → a **colder** mountain breeze.

Grass keeps moisture and gives off water vapour, so its temperature changes less → a **milder (less cold), moister** breeze.

6 **You have witnessed weather phenomena, such as winds, storms, rainfall, etc. Which atmospheric layer is mainly responsible for such phenomena and what is the primary reason for its occurrence?**

✔ ANSWER

Troposphere (0–12 km).

It is heated from the Earth's surface, so temperature falls with height (~6.5 °C/km). Warm air near the ground rises, cools and forms clouds and rain; this rising and mixing of air drives winds and storms. Almost all water vapour is in this layer.

The main cause is **uneven heating of the Earth's surface by the Sun**.

7 **Explain the processes involved in the nitrogen cycle. How would life on Earth be affected if nitrogen were not cycled?**

✔ ANSWER

- **Nitrogen fixation:** Rhizobium (root nodules of legumes) and Azotobacter (soil) change atmospheric N_2 into ammonia (NH_3); lightning also fixes some nitrogen.

- **Nitrification:** Nitrosomonas changes ammonia to nitrite (NO_2^-); Nitrobacter changes nitrite to nitrate (NO_3^-).
- **Assimilation:** plants absorb these compounds from the soil to make proteins and nucleic acids; animals get nitrogen by eating plants/animals.
- **Ammonification:** decomposers (bacteria, fungi) break down dead bodies and wastes and return ammonia to the soil.
- **Denitrification:** Pseudomonas changes some nitrates back to N_2 gas.

If nitrogen were not cycled, plants could not get usable nitrogen, so they could not make proteins and DNA; plant growth would stop, and animals and humans that depend on plants would not survive.

8 **What are the impacts of deforestation on the Earth's oxygen and carbon cycles? What are the other consequences of deforestation?**

✔ ANSWER

Oxygen and carbon cycles:

- Less photosynthesis → less O_2 released and less CO_2 absorbed; trees that are burnt or rot release their carbon → **CO_2 in the air rises**, increasing global warming.

Other consequences:

- Less transpiration → decline in local rainfall.
- Change in surface albedo.
- No roots to hold soil → soil erosion, floods.
- Loss of habitats → decline in biodiversity.

9 **Explain with a suitable diagram the path that carbon takes to go back to the atmosphere. You may start from plants using CO_2 from the atmosphere.**

✔ ANSWER

Diagram (draw as a cycle):

CO_2 in air → (photosynthesis) → glucose in plants → (eaten) → animals

Plants and animals → (respiration) → CO_2 in air

Dead plants and animals → (decomposition) → CO_2 in air

Dead remains buried for millions of years → fossil fuels (coal, oil, gas) → (combustion) → CO_2 in air

Explanation: plants use CO_2 to make glucose. Carbon returns to the air quickly by respiration of plants and animals and by decomposition of dead bodies (fast cycle). Some remains become fossil fuels over millions of years; burning them releases CO_2 (slow cycle). The ocean also exchanges CO_2 with the air.

10 **Why is an excess of CO_2 in the atmosphere considered undesirable even though it is required by plants?**

✔ ANSWER

Plants need only a balanced amount of CO_2 for photosynthesis.

Excess CO_2 traps more heat (stronger greenhouse effect) → **global warming**, melting glaciers and Arctic ice, rising sea level and more extreme weather.

It also makes the ocean more acidic, harming plankton and coral reefs. Since 1960 CO_2 has risen about 35% (315 → 420 ppm).

11 **How is heat lost from the surface of the Earth? What is its significance?**

💡 SAMPLE ANSWER

The surface absorbs sunlight and **re-radiates heat as infrared radiation** into the atmosphere; warm air also rises and carries heat upward.

Significance: part of this outgoing heat is trapped by greenhouse gases, keeping the Earth warm enough for life; the rest escapes to space, so the Earth does not keep heating up. This balance keeps the temperature suitable. Re-radiation also explains why concrete houses and cities stay hot at night.

12 **If the Earth were a flat disc instead of a sphere, how would the patterns of solar radiation and temperature be different?**

💡 SAMPLE ANSWER

If the flat face faced the Sun, rays would strike every place at nearly the same angle.

So sunlight would be spread **almost evenly** and temperature would be nearly the same everywhere: no hot equator and cold poles.

With little temperature difference there would be weak pressure belts, so planetary winds and ocean currents would be much weaker.

(The edge of the disc would get very little light.)

13 **Suppose there is a rise in atmospheric temperature on Earth. How would this affect the cryosphere, hydrosphere and biosphere?**

✔ ANSWER

- **Cryosphere:** glaciers and polar ice melt faster.

- **Hydrosphere:** rivers get more water first, sea level rises, low-lying and coastal areas flood; rainfall becomes uneven (floods and droughts); oceans become warmer and absorb less CO_2.
- **Biosphere:** habitats are lost; crops and fisheries are affected; plankton and coral reefs are harmed; biodiversity falls.

14 **Explain how the Earth's atmosphere helps in maintaining a suitable temperature for life to survive on the Earth.**

✔ ANSWER

- It partly absorbs incoming sunlight: the ozone layer blocks harmful UV rays, clouds and gases absorb or reflect some sunlight, so the surface does not get too hot in the day.
- It traps outgoing heat: greenhouse gases (CO_2, CH_4, water vapour) absorb the infrared heat re-radiated by the surface and stop it from escaping to space, so nights are not too cold.

Without the atmosphere the Earth would be too cold for life.

15 **Describe the interrelationship between different spheres of the Earth. Illustrate with example how these spheres function in a delicate balance.**

✔ ANSWER

The five spheres — geosphere (rocks, soil), hydrosphere (water), cryosphere (ice, snow), atmosphere (air) and biosphere (living things) — constantly exchange energy and matter through sunlight, winds, water and nutrient cycles.

Example: the Sun melts snow (cryosphere) → water fills the lake (hydrosphere) → it waters the soil (geosphere) → grass grows and feeds sheep (biosphere); water evaporates into the air (atmosphere) and falls again as snow.

A disturbance in one spreads to the others: warmer air (atmosphere) melts glaciers (cryosphere) → sea level rises (hydrosphere) → coasts flood (geosphere) → habitats are lost (biosphere).

⭐ The Journey Beyond

1 **Two Earth-sized planets with an atmosphere: one fully covered by ocean, the other fully land. How would their wind patterns compare with the winds on Earth?**

💡 SAMPLE ANSWER

Both planets would still have pressure belts (equator low, ~30° high, ~60° low, poles high) because the Sun heats the equator more; rotation would still curve the winds.

- **Ocean planet:** water heats and cools slowly, so temperature differences are smaller → steady, regular, weaker winds in neat belts; no land/sea breezes.
- **Land planet:** land heats and cools fast → large day–night and equator–pole differences → stronger, more variable winds; dry, with few clouds or rain.
- **Earth:** mixed land and sea break the belts and add monsoons, land and sea breezes and local winds.

2 **Choose one meal you ate recently. For each main item explain (i) how its carbon came from CO_2 in the air, (ii) how its nitrogen came from the atmosphere**

into the soil and the plant. List other activities in producing/cooking the meal that add CO_2 or nitrogen to the environment.

💡 **SAMPLE ANSWER**

Example: roti and dal.

(i) Wheat (roti) and pulse (dal) plants took CO_2 from the air and made glucose by photosynthesis; it was stored as starch and proteins in the grains.

(ii) Dal: Rhizobium in the root nodules of the pulse plant fixed N_2 from the air. Wheat: nitrogen came from soil nitrates made by bacteria and from urea fertiliser made by the Haber-Bosch process; roots absorbed nitrates to make proteins.

Activities adding CO_2 / nitrogen: diesel tractors and pumps, making fertilisers, excess fertiliser washing into water, transport by trucks, milling with electricity, cooking on LPG/wood, packaging and food waste.

3 **Using IMD (https://mausam.imd.gov.in/) or newspaper data, compare average monsoon rainfall (June–September) of your city/district for 5 years in two decades (e.g. 1980s and 2020s); note the trend in heavy rain days (>50 mm). Connect it to warmer Arabian Sea or land-use change.**

💡 **SAMPLE ANSWER**

Make a table: Year | June–Sept rainfall (mm) | Days with >50 mm rain; find the average for each decade.

Likely finding in many places: total rainfall changes a little, but **days of very heavy rain increase** and dry gaps get longer.

Link: a warmer Arabian Sea gives more evaporation and moisture, so rain comes in heavy bursts; cutting forests and building cities reduces infiltration and transpiration, increasing run-off and local heat.

⭐ The Quest Continues ...

? What new discoveries will new tools (that observe the planet in real time) reveal about the changing Earth, and how will they improve our understanding of global warming and climate change?

💡 **SAMPLE ANSWER**

Satellites, ocean buoys, weather stations and computer models give real-time data on temperature, ice, sea level, forests, CO_2 and rainfall.

They can show how fast glaciers are melting, where forests are lost, how monsoons are changing and how the spheres affect one another.

This helps give earlier warnings of floods, droughts and cyclones, and better forecasts of climate change, so we can plan to reduce harm.

🎉 **Chapter 13 done! Revise the answers once more tomorrow.**